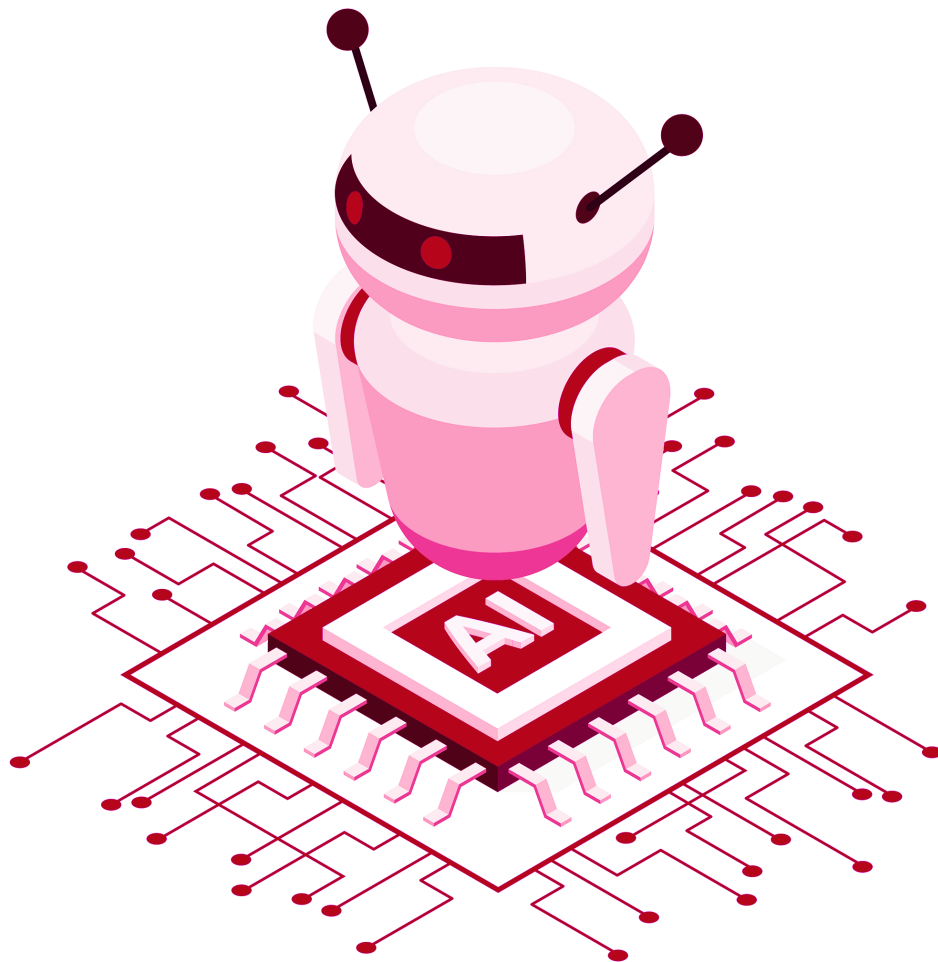




Deep Reinforcement Learning

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Homework 9:

Exploration Methods

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1 Light-tailed Distributions[25-points]

1.1 Hoeffding's Inequality[10-points]

1.1.1 a)[6-points]

For a random variable X with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$, we prove that for any $s > 0$,

$$\mathbb{E}[e^{sX}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right).$$

Proof. Assume $a < b$ (the equality case is trivial). Since $\mathbb{E}[X] = 0$ and $a \leq X \leq b$, we must have $a < 0 < b$. Consider the two-point distribution:

$$P(X' = a) = \frac{b}{b-a}, \quad P(X' = b) = \frac{-a}{b-a}.$$

This satisfies $\mathbb{E}[X'] = 0$. Let $D = b - a > 0$ and $w = -a/D$ ($0 < w < 1$), so $a = -wD$, $b = (1-w)D$. The MGF is:

$$\mathbb{E}[e^{sX'}] = (1-w)e^{-swD} + we^{s(1-w)D}.$$

Set $t = sD$ and define:

$$k(t) = \ln((1-w)e^{-wt} + we^{(1-w)t}).$$

Then $k(0) = 0$, $k'(0) = 0$, and the second derivative satisfies $k''(t) \leq \frac{1}{4}$. By Taylor's theorem:

$$k(t) = \frac{1}{2}k''(\xi)t^2 \leq \frac{t^2}{8} = \frac{s^2D^2}{8} = \frac{s^2(b-a)^2}{8}.$$

Thus $\mathbb{E}[e^{sX'}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right)$. For general X , this bound holds due to convexity of the exponential function. $\square$

1.1.2 b)[4-points]

Let $Z_1, \dots, Z_n$ be independent random variables with $Z_i \in [a, b]$ for all i , where $-\infty < a \leq b < \infty$. For all $t \geq 0$:

$$\begin{aligned} \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n (Z_i - \mathbb{E}[Z_i]) \geq t\right) &\leq \exp\left(\frac{-2nt^2}{(b-a)^2}\right), \\ \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n (Z_i - \mathbb{E}[Z_i]) \leq -t\right) &\leq \exp\left(\frac{-2nt^2}{(b-a)^2}\right). \end{aligned}$$

Proof. Define $X_i = Z_i - \mathbb{E}[Z_i]$, so $\mathbb{E}[X_i] = 0$ and $X_i \in [a - \mathbb{E}[Z_i], b - \mathbb{E}[Z_i]]$ with range $b - a$. By part (a):

$$\mathbb{E}[e^{sX_i}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right) \quad \forall s > 0.$$

Let $S_n = \sum_{i=1}^n X_i$. By independence:

$$\mathbb{E}[e^{sS_n}] \leq \exp\left(\frac{ns^2(b-a)^2}{8}\right).$$

For the upper tail ($S_n \geq nt$):

$$\mathbb{P}(S_n \geq nt) \leq \inf_{s>0} e^{-snt} \mathbb{E}[e^{sS_n}] \leq \inf_{s>0} \exp\left(-snt + \frac{ns^2(b-a)^2}{8}\right).$$

Minimizing the exponent: set derivative w.r.t s to zero:

$$-nt + \frac{ns(b-a)^2}{4} = 0 \implies s = \frac{4t}{(b-a)^2}.$$

Substituting gives:

$$\mathbb{P}(S_n \geq nt) \leq \exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

For the lower tail, apply the same argument to $-S_n$, yielding identical bound. □

1.2 Sub-Gaussian[15-points]

- (1) Upper tail: $\Pr(X > \mu + t) \leq e^{-t^2/(2\sigma^2)}$ for $t > 0$
- (2) Lower tail: $\Pr(X < \mu - t) \leq e^{-t^2/(2\sigma^2)}$ for $t > 0$
- (3) Two-sided tail: $\Pr(|X - \mu| \geq t) \leq 2e^{-t^2/(2\sigma^2)}$ for $t > 0$

1.2.1 a-1)[2-points]

Proof. For any $\lambda > 0$, by Markov's inequality:

$$\Pr(X > \mu + t) = \Pr(e^{\lambda(X-\mu)} > e^{\lambda t}) \leq \frac{\mathbb{E}[e^{\lambda(X-\mu)}]}{e^{\lambda t}} \leq e^{\lambda^2 \sigma^2 / 2 - \lambda t}.$$

Minimize the exponent $\lambda^2 \sigma^2 / 2 - \lambda t$ over $\lambda > 0$. Set derivative to zero:

$$\frac{d}{d\lambda} \left(\frac{\lambda^2 \sigma^2}{2} - \lambda t \right) = \lambda \sigma^2 - t = 0 \implies \lambda = \frac{t}{\sigma^2}.$$

Substituting yields:

$$\Pr(X > \mu + t) \leq \exp\left(\frac{(t/\sigma^2)^2 \sigma^2}{2} - \frac{t}{\sigma^2} \cdot t\right) = \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

□

1.2.2 a-2)[2-points]

Proof. For any $\lambda < 0$, set $\eta = -\lambda > 0$. Then:

$$\Pr(X < \mu - t) = \Pr(e^{\lambda(X-\mu)} > e^{\lambda(-t)}) = \Pr(e^{-\eta(X-\mu)} > e^{\eta t}) \leq \frac{\mathbb{E}[e^{-\eta(X-\mu)}]}{e^{\eta t}} \leq e^{\eta^2 \sigma^2 / 2 - \eta t}.$$

Minimize $\eta^2 \sigma^2 / 2 - \eta t$ over $\eta > 0$. The minimum occurs at $\eta = t / \sigma^2$:

$$\Pr(X < \mu - t) \leq \exp\left(\frac{(t/\sigma^2)^2 \sigma^2}{2} - \frac{t}{\sigma^2} \cdot t\right) = \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

□

1.2.3 a-3)[2-points]

Proof.

$$\Pr(|X - \mu| \geq t) = \Pr(X \geq \mu + t) + \Pr(X \leq \mu - t) \leq e^{-t^2/(2\sigma^2)} + e^{-t^2/(2\sigma^2)} = 2e^{-t^2/(2\sigma^2)}.$$

□

1.2.4 b)[3-points]

Suppose $\{X_i\}_{i=1}^n$ are independent random variables with $\mathbb{E}[X_i] = \mu_i$ and each X_i is sub-Gaussian with parameter σ_i . Then for all $t \geq 0$,

$$\Pr\left[\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right] \leq 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

Proof. Define $Y_i = X_i - \mu_i$ for each i . Then $\mathbb{E}[Y_i] = 0$ and Y_i is sub-Gaussian with parameter σ_i , satisfying:

$$\mathbb{E}[e^{\lambda Y_i}] \leq e^{\lambda^2 \sigma_i^2 / 2} \quad \forall \lambda \in \mathbb{R}.$$

Let $S_n = \sum_{i=1}^n Y_i$. By independence, the moment generating function of S_n is:

$$\mathbb{E}[e^{\lambda S_n}] = \prod_{i=1}^n \mathbb{E}[e^{\lambda Y_i}] \leq \prod_{i=1}^n e^{\lambda^2 \sigma_i^2 / 2} = \exp\left(\frac{\lambda^2}{2} \sum_{i=1}^n \sigma_i^2\right).$$

Thus S_n is sub-Gaussian with parameter $\sigma = \sqrt{\sum_{i=1}^n \sigma_i^2}$. Applying the two-sided tail bound for sub-Gaussian variables:

$$\Pr[|S_n| \geq t] \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right) = 2 \exp\left(-\frac{t^2}{2 \sum_{i=1}^n \sigma_i^2}\right).$$

□

1.2.5 c)[4-points]

Let $X_1, \dots, X_n$ be i.i.d. sub-Gaussian random variables with variance proxy σ^2 and $\mathbb{E}[X] = \mu$. Then:

1. For any $\epsilon \geq 0$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq \epsilon\right) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

2. For any $\delta \in [0, 1]$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu < \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}}\right) \geq 1 - \delta.$$

Proof. Define $Y_i = X_i - \mu$ for each i , so $\mathbb{E}[Y_i] = 0$ and Y_i is sub-Gaussian with parameter σ . Let $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$. The sum $S_n = \sum_{i=1}^n Y_i$ is sub-Gaussian with parameter $\sigma\sqrt{n}$ as shown in part (b). Thus $\bar{Y}_n = S_n/n$ is sub-Gaussian with parameter $\sigma/\sqrt{n}$:

$$\mathbb{E}[e^{\lambda \bar{Y}_n}] = \mathbb{E}[e^{\lambda S_n/n}] \leq \exp\left(\frac{\lambda^2}{2} \cdot \frac{\sigma^2}{n}\right) = \exp\left(\frac{\lambda^2 \sigma^2}{2n}\right).$$

1. Apply the upper tail bound for sub-Gaussian variables:

$$\mathbb{P}(\bar{Y}_n \geq \epsilon) \leq \inf_{\lambda > 0} e^{-\lambda \epsilon} \exp\left(\frac{\lambda^2 \sigma^2}{2n}\right) = \inf_{\lambda > 0} \exp\left(-\lambda \epsilon + \frac{\lambda^2 \sigma^2}{2n}\right).$$

Minimize the exponent: set derivative w.r.t λ to zero:

$$-\epsilon + \frac{\lambda \sigma^2}{n} = 0 \implies \lambda = \frac{n\epsilon}{\sigma^2}.$$

Substituting gives:

$$\mathbb{P}(\bar{Y}_n \geq \epsilon) \leq \exp\left(-\frac{n\epsilon}{\sigma^2} \cdot \epsilon + \frac{(n\epsilon/\sigma^2)^2 \sigma^2}{2n}\right) = \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

2. Set $\epsilon = \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}}$. Then:

$$\mathbb{P}(\bar{Y}_n \geq \epsilon) \leq \exp\left(-\frac{n}{2\sigma^2} \cdot \frac{2\sigma^2 \log(1/\delta)}{n}\right) = \exp(-\log(1/\delta)) = \delta.$$

Thus:

$$\mathbb{P}(\bar{Y}_n < \epsilon) = 1 - \mathbb{P}(\bar{Y}_n \geq \epsilon) \geq 1 - \delta.$$

Substituting the expression for ϵ gives the result.

□

2 UCB[75-points]

2.1 The Upper Confidence Bound Algorithm[40-points]

2.1.1 a)[2-points]

For any policy π and stochastic bandit environment ν with action space $\mathcal{A}$ (finite or countable) and horizon $n \in \mathbb{N}$, the regret R_n satisfies

$$R_n = \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]$$

where $\Delta_a = R_{\max} - R_a$ is the regret for arm a , $R_{\max} = \max_{a' \in \mathcal{A}} R_{a'}$ is the maximum expected reward, and $T_a(n)$ is the number of times arm a is pulled up to time n .

Proof. The regret is defined as:

$$R_n = nR_{\max} - \mathbb{E} \left[\sum_{t=1}^n r_{I_t} \right]$$

where I_t is the arm pulled at time t and r_{I_t} is the reward obtained.

Rewriting the total reward sum by grouping pulls per arm:

$$\sum_{t=1}^n r_{I_t} = \sum_{a \in \mathcal{A}} \sum_{k=1}^{T_a(n)} r_a(k)$$

where $r_a(k)$ is the reward from the k -th pull of arm a . Taking expectations:

$$\mathbb{E} \left[\sum_{t=1}^n r_{I_t} \right] = \mathbb{E} \left[\sum_{a \in \mathcal{A}} \sum_{k=1}^{T_a(n)} r_a(k) \right]$$

By the law of total expectation and the i.i.d. property of rewards for each arm:

$$\mathbb{E} \left[\sum_{k=1}^{T_a(n)} r_a(k) \right] = \mathbb{E} \left[\mathbb{E} \left[\sum_{k=1}^{T_a(n)} r_a(k) \mid T_a(n) \right] \right] = \mathbb{E}[T_a(n) R_a]$$

Thus:

$$\mathbb{E} \left[\sum_{t=1}^n r_{I_t} \right] = \sum_{a \in \mathcal{A}} \mathbb{E}[T_a(n) R_a] = \sum_{a \in \mathcal{A}} R_a \mathbb{E}[T_a(n)]$$

Substituting into the regret expression:

$$R_n = nR_{\max} - \sum_{a \in \mathcal{A}} R_a \mathbb{E}[T_a(n)]$$

Since $\sum_{a \in \mathcal{A}} T_a(n) = n$ almost surely, we have:

$$\sum_{a \in \mathcal{A}} \mathbb{E}[T_a(n)] = n$$

Rewriting:

$$R_n = R_{\max} \sum_{a \in \mathcal{A}} \mathbb{E}[T_a(n)] - \sum_{a \in \mathcal{A}} R_a \mathbb{E}[T_a(n)] = \sum_{a \in \mathcal{A}} (R_{\max} - R_a) \mathbb{E}[T_a(n)] = \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]$$

□

2.1.2 b)[4-points]

With fixed $\delta = C$, after initialization (each arm pulled once):

1. Arm 1's UCB index becomes **fixed** at:

$$\text{UCB}_1 = \hat{\mu}_1(1) + \sqrt{2 \log(1/\delta)}$$

2. In the bad event:

$$\text{UCB}_1 < \mu_1$$

3. At least one suboptimal arm j satisfies:

$$\text{UCB}_j(1) > \mu_1$$

4. For any future $T_j \geq 1$:

$$\text{UCB}_j(T_j) \geq \mu_j > \text{UCB}_1$$

Thus, arm 1 is never selected again.

Assume the bad event occurs:

$$\hat{\mu}_1(1) < \mu_1 - \sqrt{2 \log(1/\delta)} \tag{1}$$

$$\exists j \neq 1 : \hat{\mu}_j(1) + \sqrt{2 \log(1/\delta)} > \mu_1 \tag{2}$$

Let $d = \mu_1 - \text{UCB}_1 > 0$. For any suboptimal arm j :

$$\mu_j \geq \min_{k \neq 1} \mu_k > \mu_1 - d = \text{UCB}_1$$

After m pulls of arm j :

$$\text{UCB}_j(m) = \underbrace{\hat{\mu}_j(m)}_{\rightarrow \mu_j} + \underbrace{\sqrt{\frac{c^2}{m}}}_{\geq 0} > \text{UCB}_1$$

for sufficiently large m (and in practice, for all $m \geq 1$).

Algorithm Never Returns to Arm 1

At any time $t > K$:

- UCB_1 remains fixed at $L < \mu_1$
- All arms j pulled at least once have $UCB_j \geq \mu_j > L$
- The algorithm always observes:

$$\max_{j \neq 1} UCB_j(t) > UCB_1$$

- Thus, it always selects a suboptimal arm

Linear Regret Consequence

The regret grows linearly:

$$R_n \geq (n - K)\Delta_{\min}, \quad \Delta_{\min} = \min_{j \neq 1} (\mu_1 - \mu_j) > 0$$

with probability $\geq \delta^K$ (since all arms must be misestimated).

Solution: Time-Varying δ

Set $\delta_t = t^{-\alpha}$ ($\alpha > 1$). The UCB index becomes:

$$UCB_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{2\alpha \log t}{T_i(t)}}$$

This ensures:

- Confidence terms grow with $\log t$
- Optimal arm gets pulled infinitely often
- Sublinear regret: $R_n = \mathcal{O}(\log n)$

2.1.3 c)[4-points]

If event G_i occurs, then arm i is played at most u_i times: $T_i(n) \leq u_i$.

Proof. Assume G_i occurs. Suppose for contradiction that arm i is played more than u_i times. Let τ be the time of the $(u_i + 1)$ -th pull of arm i . At time $\tau - 1$, $T_i(\tau - 1) = u_i$. By G_i :

$$UCB_i(\tau - 1, \delta) = \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}} < \mu_1.$$

Also by G_i :

$$\mu_1 < \min_{t \in [n]} UCB_1(t, \delta) \leq UCB_1(\tau - 1, \delta).$$

Thus:

$$UCB_i(\tau - 1, \delta) < \mu_1 < UCB_1(\tau - 1, \delta).$$

This contradicts arm i being pulled at time τ . Hence, $T_i(n) \leq u_i$. □

2.1.4 d)[4-points]

$$\mathbb{E}[T_i(n)] \leq u_i + \mathbb{P}(G_i^c) \cdot n$$

Proof. Decompose the expectation:

$$\mathbb{E}[T_i(n)] = \mathbb{E}[T_i(n) \mid G_i] \mathbb{P}(G_i) + \mathbb{E}[T_i(n) \mid G_i^c] \mathbb{P}(G_i^c).$$

By (c), $\mathbb{E}[T_i(n) \mid G_i] \leq u_i$. Trivially, $\mathbb{E}[T_i(n) \mid G_i^c] \leq n$. Thus:

$$\mathbb{E}[T_i(n)] \leq u_i \mathbb{P}(G_i) + n \mathbb{P}(G_i^c) \leq u_i + n \mathbb{P}(G_i^c).$$

□

2.1.5 e)[6-points]

Assume u_i satisfies $\Delta_i - \sqrt{\frac{2 \log(1/\delta)}{u_i}} \geq c\Delta_i$ for $c \in (0, 1)$. Then:

$$\mathbb{P} \left(\hat{\mu}_{i,u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}} \geq \mu_1 \right) \leq \exp \left(-\frac{u_i c^2 \Delta_i^2}{2} \right).$$

Proof. The event implies:

$$\hat{\mu}_{i,u_i} - \mu_i \geq \mu_1 - \mu_i - \sqrt{\frac{2 \log(1/\delta)}{u_i}} = \Delta_i - \sqrt{\frac{2 \log(1/\delta)}{u_i}} \geq c\Delta_i.$$

Since $\hat{\mu}_{i,u_i}$ is the average of u_i independent 1-sub-Gaussian random variables:

$$\mathbb{P}(\hat{\mu}_{i,u_i} - \mu_i \geq c\Delta_i) \leq \exp \left(-\frac{u_i (c\Delta_i)^2}{2} \right) = \exp \left(-\frac{u_i c^2 \Delta_i^2}{2} \right).$$

□

2.1.6 f)[4-points]

$$\mathbb{P}(G_i^c) \leq n\delta + \exp \left(-\frac{u_i c^2 \Delta_i^2}{2} \right)$$

Proof. By union bound:

$$\mathbb{P}(G_i^c) \leq \mathbb{P} \left(\mu_1 \geq \min_{t \in [n]} \text{UCB}_1(t, \delta) \right) + \mathbb{P} \left(\hat{\mu}_{i,u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}} \geq \mu_1 \right).$$

The second term is bounded in (e). For the first term:

$$\left\{ \mu_1 \geq \min_{t \in [n]} \text{UCB}_1(t, \delta) \right\} \subseteq \bigcup_{t=1}^n \{ \mu_1 \geq \text{UCB}_1(t, \delta) \}.$$

Each event $\{\mu_1 \geq \text{UCB}_1(t, \delta)\}$ is equivalent to:

$$\hat{\mu}_{1,s} \leq \mu_1 - \sqrt{\frac{2 \log(1/\delta)}{s}} \quad (s = T_1(t)),$$

which has probability $\leq \delta$ by sub-Gaussian tail. Union over $s = 1$ to n gives:

$$\mathbb{P} \left(\bigcup_{t=1}^n \{\mu_1 \geq \text{UCB}_1(t, \delta)\} \right) \leq \sum_{s=1}^n \delta = n\delta.$$

□

2.1.7 g)[6-points]

With appropriate u_i and $c \in (0, 1)$:

$$\mathbb{E}[T_i(n)] \leq 3 + \frac{16 \log n}{\Delta_i^2}.$$

Proof. From (d) and (f):

$$\mathbb{E}[T_i(n)] \leq u_i + n \left[n\delta + \exp \left(-\frac{u_i c^2 \Delta_i^2}{2} \right) \right].$$

Set $\delta = \frac{1}{n^2}$ and $u_i = \frac{16 \log n}{\Delta_i^2}$. Then $n^2 \delta = 1$. Choose $c = \frac{1}{2}$. Now:

$$n \exp \left(-\frac{u_i c^2 \Delta_i^2}{2} \right) = n \exp \left(-\frac{16 \log n \cdot (1/4) \Delta_i^2}{2 \Delta_i^2} \right) = n \exp(-2 \log n) = n^{-1}.$$

Thus:

$$\mathbb{E}[T_i(n)] \leq \frac{16 \log n}{\Delta_i^2} + 1 + 1 = \frac{16 \log n}{\Delta_i^2} + 2 \leq 3 + \frac{16 \log n}{\Delta_i^2}.$$

□

2.1.8 h)[5-points]

If $\delta = \frac{1}{n^2}$:

$$R_n \leq 3 \sum_{i=1}^k \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log n}{\Delta_i}.$$

Proof. By regret decomposition:

$$R_n = \sum_{i=1}^k \Delta_i \mathbb{E}[T_i(n)].$$

For optimal arm ($\Delta_i = 0$), contribution is 0. For suboptimal arms ($\Delta_i > 0$), apply (g):

$$R_n \leq \sum_{i: \Delta_i > 0} \Delta_i \left(3 + \frac{16 \log n}{\Delta_i^2} \right) = 3 \sum_{i: \Delta_i > 0} \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log n}{\Delta_i}.$$

Extending sums to all i (optimal arm adds 0) gives the result.

□

2.1.9 i)[5-points]

$$R_n \leq 8\sqrt{nk \log n} + 3 \sum_{i=1}^k \Delta_i.$$

Proof. Set $\Delta = \sqrt{\frac{16k \log n}{n}}$. Split arms:

- $\Delta_i \geq \Delta$: Use (h) directly
- $0 < \Delta_i < \Delta$: Trivial bound $\Delta_i \mathbb{E}[T_i(n)] \leq \Delta n$

Then:

$$R_n = \sum_{i: \Delta_i \geq \Delta} \Delta_i \mathbb{E}[T_i(n)] + \sum_{i: 0 < \Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)].$$

First sum:

$$\sum_{i: \Delta_i \geq \Delta} \Delta_i \mathbb{E}[T_i(n)] \leq \sum_{i: \Delta_i \geq \Delta} \left(3\Delta_i + \frac{16 \log n}{\Delta_i} \right) \leq 3 \sum_{i=1}^k \Delta_i + \sum_{i: \Delta_i \geq \Delta} \frac{16 \log n}{\Delta_i}.$$

Bounding the sum:

$$\sum_{i: \Delta_i \geq \Delta} \frac{16 \log n}{\Delta_i} \leq k \cdot \frac{16 \log n}{\Delta} = 16k \log n \cdot \sqrt{\frac{n}{16k \log n}} = 4\sqrt{nk \log n}.$$

Second sum:

$$\sum_{i: 0 < \Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)] \leq \Delta \sum_i \mathbb{E}[T_i(n)] = \Delta \cdot n = n \cdot \sqrt{\frac{16k \log n}{n}} = 4\sqrt{nk \log n}.$$

Combining:

$$R_n \leq 3 \sum_{i=1}^k \Delta_i + 4\sqrt{nk \log n} + 4\sqrt{nk \log n} = 3 \sum_{i=1}^k \Delta_i + 8\sqrt{nk \log n}.$$

□

2.2 Power of 2 version of UCB Algorithm^{*}(*Bonus*)[35 – points]

Regret Analysis

Define $\mu^* = \max_i \mu_i$ and $\Delta_i = \mu^* - \mu_i$. Let $\delta = \frac{1}{Kn^2}$ and $L = \lfloor \log_2(n) \rfloor$ be the number of phases. The total regret is bounded by considering two events: the good event G where all confidence bounds hold, and the bad event G^c .

Good Event Definition

Define the good event:

$$G = \bigcap_{i=1}^K \bigcap_{\ell=0}^L \left\{ \hat{\mu}_i(t_\ell) - \mu_i \leq \sqrt{\frac{2 \log(1/\delta)}{T_i(t_\ell)}} \right\}$$

where t_ℓ is the start time of phase ℓ . By union bound and sub-Gaussian tail:

$$\mathbb{P}(G^c) \leq \sum_{i=1}^K \sum_{\ell=0}^L \delta \leq K(L+1)\delta \leq K \log_2(n) \cdot \frac{1}{Kn^2} \leq \frac{\log_2 n}{n^2} \leq \frac{1}{n}$$

Regret under Good Event

Under G , if arm i has been pulled $T_i \geq \frac{8 \log(1/\delta)}{\Delta_i^2}$ times, then:

$$\text{UCB}_i \leq \mu_i + \sqrt{\frac{2 \log(1/\delta)}{T_i}} \leq \mu_i + \frac{\Delta_i}{2} = \mu^* - \frac{\Delta_i}{2} < \mu^* \leq \text{UCB}^*$$

Thus, arm i is not selected once $T_i \geq u_i \triangleq \frac{8 \log(1/\delta)}{\Delta_i^2}$.

The total pulls of arm i satisfy:

$$T_i(n) \leq 1 + u_i + 2^{m_i} \quad \text{where} \quad 2^{m_i} \leq u_i$$

Thus:

$$T_i(n) \leq 1 + u_i + u_i = 1 + 2u_i$$

The regret from suboptimal arms:

$$\sum_{i: \Delta_i > 0} \Delta_i T_i(n) \leq \sum_{i: \Delta_i > 0} \Delta_i (1 + 2u_i) = \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{16 \log(1/\delta)}{\Delta_i} \right)$$

Regret under Bad Event

The bad event contributes at most:

$$\mathbb{E}[R_n | G^c] \mathbb{P}(G^c) \leq n \Delta_{\max} \cdot \frac{1}{n} = \Delta_{\max}$$

Total Expected Regret

Combining both cases:

$$\mathbb{E}[R_n] \leq \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{16 \log(1/\delta)}{\Delta_i} \right) + \Delta_{\max}$$

Substituting $\delta = \frac{1}{Kn^2}$ and $\log(1/\delta) = \log(Kn^2) = \log K + 2 \log n$:

$$\mathbb{E}[R_n] \leq \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{16(\log K + 2 \log n)}{\Delta_i} \right) + \Delta_{\max}$$

Minimax Regret Bound

To derive a minimax bound, set $\Delta = \sqrt{\frac{16K(\log K + 2\log n)}{n}}$. Split arms:

$$\begin{aligned}
 \mathbb{E}[R_n] &\leq \sum_{i:\Delta_i \geq \Delta} \left(\Delta_i + \frac{16(\log K + 2\log n)}{\Delta_i} \right) + \sum_{i:0 < \Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)] + \Delta_{\max} \\
 &\leq \sum_{i=1}^K \Delta_i + 16(\log K + 2\log n) \cdot \frac{K}{\Delta} + 2n\Delta \\
 &= \sum_{i=1}^K \Delta_i + 4\sqrt{nK(\log K + 2\log n)} + 8\sqrt{nK(\log K + 2\log n)} \\
 &\leq \sum_{i=1}^K \Delta_i + 12\sqrt{nK(\log K + 2\log n)}
 \end{aligned}$$

Since $\sum_{i=1}^K \Delta_i \leq K\Delta_{\max} = O(K)$, the minimax regret is:

$$\mathbb{E}[R_n] = O\left(\sqrt{nK(\log K + 2\log n)}\right)$$

3 Online Learning[50-points]

3.1 Randomized Weighted Majority Algorithm[35-points]

3.1.1 a)[5-points]

For any time $t \geq 0$,

$$\mathbb{E}[S_{t+1}] = \mathbb{E}[S_t] \cdot (1 - \epsilon \cdot \mathbb{P}(\hat{m}_t = 1))$$

Proof. The weight update rule is:

$$w_i(t+1) = w_i(t) \cdot (1 - \epsilon \cdot m_i(t))$$

where $m_i(t) = 1$ if expert i is incorrect at time t , else 0. The total weight is:

$$S_{t+1} = \sum_{i=1}^N w_i(t)(1 - \epsilon \cdot m_i(t)) = S_t - \epsilon \sum_{i=1}^N w_i(t)m_i(t).$$

Let $\eta_t = \sum_{i=1}^N \frac{w_i(t)}{S_t} m_i(t)$. Then:

$$\sum_{i=1}^N w_i(t)m_i(t) = S_t \eta_t,$$

and we can say:

$$\hat{m}_t \sim \text{Bern}(\mathbb{P}(\hat{m}_t = 1)) \implies \mathbb{P}(\hat{m}_t = 1) = \mathbb{E}[\hat{m}_t] = \sum_{i=1}^N \frac{w_i(t)}{S_t} \cdot m_i(t) = \eta_t$$

so

$$S_{t+1} = S_t - \epsilon S_t \eta_t = S_t(1 - \epsilon \cdot \mathbb{P}(\hat{m}_t = 1)).$$

Taking expectation gives the result:

$$\mathbb{E}[S_{t+1}] = \mathbb{E}[S_t] \cdot (1 - \epsilon \cdot \mathbb{P}(\hat{m}_t = 1)).$$

□

3.1.2 b)[8-points]

After T rounds,

$$\mathbb{E}[S_T] \leq N \cdot \exp \left(-\epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1) \right).$$

Proof. From (a), $\mathbb{E}[S_{t+1}] = \mathbb{E}[S_t](1 - \epsilon\mathbb{P}(\hat{m}_t = 1)) \leq \mathbb{E}[S_t] \exp(-\epsilon\mathbb{P}(\hat{m}_t = 1))$ since $1 - x \leq e^{-x}$ for $x \in [0, 1]$. Iterating:

$$\mathbb{E}[S_T] \leq \mathbb{E}[S_0] \exp\left(-\epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1)\right) = N \exp\left(-\epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1)\right).$$

Thus the result holds. $\square$

3.1.3 c)[15-points]

For any expert i ,

$$\mathbb{E}[M] \leq (1 + \epsilon)M_i + \frac{\ln N}{\epsilon},$$

where $M = \sum_{t=0}^{T-1} \hat{m}_t$ is the algorithm's total mistakes, and M_i is expert i 's mistakes.

Proof. For any expert i :

$$\mathbb{E}[w_i(T)] = (1 - \epsilon)^{\mathbb{E}[M_i]} \leq \mathbb{E}[S_T] \leq N \exp\left(-\epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1)\right).$$

Taking logarithms and using $\ln(1 - \epsilon) \geq -\epsilon - \epsilon^2$ for $\epsilon \in [0, 0.5]$:

$$\begin{aligned} \mathbb{E}[M_i] \ln(1 - \epsilon) &\leq \ln N - \epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1) \\ \implies \mathbb{E}[M_i](-\epsilon - \epsilon^2) &\leq \ln N - \epsilon \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1) \\ \implies \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1) &\leq (1 + \epsilon)\mathbb{E}[M_i] + \frac{\ln N}{\epsilon}. \end{aligned}$$

Thus we have:

$$\mathbb{E}[M] = \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{I}(\hat{m}_t = 1)\right] = \sum_{t=0}^{T-1} \mathbb{P}(\hat{m}_t = 1) \leq (1 + \epsilon)M_i + \frac{\ln N}{\epsilon}.$$

$\square$

3.1.4 d)[7-points]

$$\mathbb{E}[M] \leq \min_{i \in [N]} M_i + 2\sqrt{T \ln N}.$$

Proof. From (c):

$$\mathbb{E}[M] \leq (1 + \epsilon) \min_i M_i + \frac{\ln N}{\epsilon}.$$

Let $R = \mathbb{E}[M] - \min_i M_i$. Then:

$$R \leq \epsilon \min_i M_i + \frac{\ln N}{\epsilon} \leq \epsilon T + \frac{\ln N}{\epsilon}.$$

Choose $\epsilon = \sqrt{\ln N/T}$ (assuming $T \geq \ln N$):

$$R \leq \sqrt{T \ln N} + \sqrt{T \ln N} = 2\sqrt{T \ln N}.$$

Thus $\mathbb{E}[M] \leq \min_i M_i + 2\sqrt{T \ln N}$.

□

The $2\sqrt{T \ln N}$ term grows sublinearly, ensuring the average regret per round $\frac{\mathbb{E}[M] - \min_i M_i}{T} \rightarrow 0$ as $T \rightarrow \infty$.

This bound matches the information-theoretic lower bound $\Omega(\sqrt{T \log N})$ for adversarial settings, making it optimal in its dependence on T and N .

The logarithmic dependence on N allows handling large expert sets while maintaining low regret.

3.2 Hedge Algorithm* (Bonus)[15 – points]

3.2.1 a)[6-points]

3.2.2 b)[7-points]

3.2.3 c)[2-points]
